

Executive Summary: HR Employee Attrition Analysis

IBM HR Analytics Dataset | Python · Pandas · Matplotlib · Seaborn

1. Data Preparation & Validation

The dataset was examined for missing values, incorrect data types, and duplicate records.

- No missing values or duplicates were found in the dataset.
- Three columns (EmployeeCount, Over18, StandardHours) were dropped as they added no value.
- After cleaning, the dataset was confirmed consistent and ready for analysis.

2. Overall Attrition Overview

- Around 16% of employees have left the company — roughly 1 in 6.
- Industry standard sits at 10–12%, meaning this company is above average and warrants a closer look.

3. Department & Job Role Analysis

Attrition varies significantly across departments and roles.

- Sales has the highest attrition rate among all departments, making it the most alarming from an HR standpoint.
- R&D follows closely, but given its larger headcount, the overall count of exits is also substantial.
- At the job role level, Sales Representatives show one of the highest attrition rates, consistent with the department-level finding.
- Lab Technicians also show high attrition counts, aligning with the R&D department trend.

4. Salary & Compensation

- Employees who left were paid roughly \$2,000 less per month on average compared to those who stayed.
- Lower income brackets show a clear concentration of attrition, suggesting compensation is a key driver.

5. Work-Life Balance, Overtime & Job Satisfaction

- Employees working overtime are about 20% more likely to leave than those who are not.
- Employees with a 'Bad' work-life balance rating are twice as likely to attrition compared to all other groups.
- Low job satisfaction correlates with a ~6% higher attrition rate than higher satisfaction tiers.

6. Age, Tenure & Promotion Insights

Several patterns emerged around employee lifecycle and career progression.

- The 18–25 age group has by far the highest attrition rate — younger employees appear to be leaving for better growth or pay opportunities.
- Average age of employees who left is noticeably lower than those who stayed, reinforcing this pattern.
- Employees at both ends of the tenure spectrum show elevated attrition — very new hires and those with 30+ years at the company.
- Employees who have not been promoted in over 6 years show among the highest attrition rates, pointing to stalled career progression as a risk factor.

7. Key Business Recommendations

1. Address overtime — reduce excessive workloads, especially in Sales and R&D where attrition is already high.
2. Review pay for high-risk groups — close the ~\$2,000 income gap for roles and departments seeing the most exits.
3. Build retention programs for young employees — structured growth paths, mentoring, and clearer timelines for progression.
4. Fix promotion delay — employees going 6+ years without a promotion are leaving. Regular review cycles could help.
5. Improve engagement and work-life balance — targeted interventions for employees flagged with low satisfaction or bad WLB scores.

8. Conclusion

Attrition in this organization is not random. It is concentrated among younger employees, lower-paid roles, sales-facing functions, and those who are overworked or under-promoted. Addressing these structural issues through targeted compensation reviews, career development, and workload management can meaningfully reduce attrition and improve long-term retention.